Hermes: Bridging Relational and Algebraic Abstractions in Homomorphically Encrypted Databases

Anonymous Author(s)

Abstract

Fully Homomorphic Encryption (FHE) promises the ability to compute over encrypted data without revealing sensitive contents. Yet, integrating it into real-world relational databases remains elusive due to prohibitive performance overhead and the structural mismatch between mutable database records and static ciphertexts. This paper presents Hermes, a system that enables homomorphically encrypted vectorized relational queries directly inside a standard SQL engine. To bridge the relational and algebraic abstractions, Hermes introduces a SIMD-aware data model that packs multiple records per ciphertext. By embedding precomputed aggregate statistics alongside data slots, the system supports efficient rotation-free aggregations. Furthermore, to overcome ciphertext immutability, we develop data-oblivious homomorphic algorithms based on slot masking and shifting, enabling secure in-place record modifications. Hermes is implemented as native loadable functions in MySQL, requiring no changes to the query planner or execution engine. Extensive evaluations on diverse datasets demonstrate up to 1600x higher throughput in encryption and over 30x speedup in insertion compared to conventional FHE implementations.

CCS Concepts

• **Security and privacy** → **Cryptography**; • **Information systems** → **Database management systems**.

Keywords

Database security, Fully homomorphic encryption, User-defined functions

ACM Reference Format:
Anonymous Author(s). 2026. Hermes: Bridging Relational and Algebraic Abstractions in Homomorphically Encrypted Databases. In . ACM, New York, NY, USA, 16 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

1.1 Background and Motivation

For over four decades, database systems have formed the backbone of modern data infrastructure, enabling structured storage, efficient retrieval, and scalable analytics. As the industry has transitioned to cloud-first architectures, the confidentiality of outsourced data has become a central concern. Increasingly, users demand end-to-end cryptographic protection throughout the entire computation pipeline, rather than merely during transmission or at rest. Among cryptographic primitives, Fully Homomorphic Encryption (FHE) offers a mechanism to evaluate algebraic operations directly on encrypted data, removing the requirement to trust the compute server. Following Gentry's seminal construction [16], subsequent schemes such as BFV and CKKS [10, 15] have improved computational efficiency and implementation maturity [3]. FHE is now applied in practical domains, including secure machine learning and privacy-preserving analytics [7, 20].

However, relational database systems have yet to fully integrate native homomorphic encryption. Although extensive research exists on encrypted databases [28, 29, 32], current practical systems frequently leak access patterns through auxiliary indices or transfer the computational workload back to the client. Operating a relational engine directly on semantically secure ciphertexts remains an open challenge. This limitation stems from a structural mismatch between database semantics and homomorphic computation. Traditional database management systems are designed for mutable records, evolving schemas, and runtime query execution plans. In contrast, FHE schemes operate over static circuits and immutable ciphertexts. Resolving this structural divide requires more than wrapping FHE function calls within SQL. It necessitates a system architecture that maps the operational dynamics of relational execution to the algebraic invariants of encrypted computation. Without bridging these abstractions, applying cryptographic schemes to secure query processing remains impractical.

Given the availability of deployable FHE toolchains and the rising expectation for secure cloud analytics, the focus of system design is determining how relational databases can integrate homomorphic execution through a native interface. This work presents a principled step toward answering that question.

1.2 Proposed Work

We present *Hermes*, a database subsystem that enables secure and efficient vectorized relational query execution over homomorphically encrypted data. Built atop the multi-slot capabilities of modern FHE schemes, Hermes is implemented as a suite of native MySQL loadable functions interfacing with OpenFHE [3]. It addresses the structural mismatch between mutable relational records and static ciphertexts by contributing three core components:

(1) SIMD-aware Relational Data Model. Hermes adopts a packed ciphertext representation that maps multiple database tuples into parallel homomorphic slots. To accelerate homomorphic aggregation, the system reserves a specific slot within each ciphertext to store a precomputed auxiliary local sum over the data slots. This value is computed at packing time and maintained incrementally. By embedding this analytic state directly into the physical layout, encrypted aggregation queries such as `SUM` can be answered using

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
Conference'17, Washington, DC, USA
© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

constant-time ciphertext additions rather than expensive Galois rotations. This layout amortizes encryption overhead and improves both throughput and storage efficiency.

(2) Data-Oblivious Mutability Primitives. To overcome ciphertext immutability, Hermes supports dynamic record-level modifications directly on packed ciphertexts. Insertions and deletions are executed via homomorphic shift-and-mask operations. Inserting a new record shifts a tail segment rightward, while deleting a record shifts the segment leftward to clear the target slot. These structural updates are performed using homomorphic masking and slot-wise addition without requiring decryption or plaintext recovery. Such operations preserve semantic security and enable efficient in-place mutability while keeping the database access patterns completely opaque to the server.

(3) FHE-Native Query Engine Integration. Hermes facilitates a full-stack encrypted query execution pipeline without modifying the core MySQL engine or query planner. The system is realized as a collection of C++ modules compiled against OpenFHE and registered via the MySQL user-defined function interface. During ingestion, records are packed into ciphertexts with embedded auxiliary sums and stored as string-encoded blobs. At query time, the system executes relational operators entirely within the encrypted domain. This architecture avoids client-side orchestration and minimizes ciphertext transformations, delivering a performant and secure relational execution engine natively within the database management system.

In Summary, this paper makes the following contributions:

- We formalize a vectorized ciphertext encoding that embeds precomputed aggregate statistics, reducing homomorphic aggregation from costly Galois rotations to constant-time additions and amortizing overall encryption overhead.
- We develop homomorphic shift-and-mask algorithms to achieve in-place slot-level insertions and deletions, enabling dynamic record mutability while preserving strict semantic security.
- We implement Hermes as a native MySQL extension to execute homomorphic query pipelines entirely within the DBMS. Extensive evaluations demonstrate up to 1,600x higher encryption throughput and 30x insertion speedup compared to conventional FHE implementations.

2 Related Work and Preliminaries

2.1 Encrypted Databases

One major class of encrypted databases relies on client-side processing, where query execution is deferred until the encrypted data is downloaded to a trusted local environment [1]. While this architecture strictly minimizes trust assumptions by keeping all analytical logic away from the cloud provider, it inherently sacrifices the computational benefits of server-side scalability. Transferring large volumes of encrypted records across the network incurs severe bandwidth penalties and high latency, effectively reducing the cloud database management system to a passive storage repository rather than an active execution engine.

An alternative approach attempts to split query execution between the client and the server, as seen in earlier information-hiding systems [19]. In this model, server-side filters retrieve encrypted tuples guided by secure indices or cryptographic tokens generated by the client. Building upon this paradigm, systems like CryptDB [29], Arx [28], and Symmetria [30] explore different structural trade-offs between query functionality and data leakage. They employ layered encryption onions, partitioned data encodings, or specialized cryptographic circuits to evaluate relational predicates. However, these designs frequently leak access patterns or require multiple rounds of network interaction, as the server must often return intermediate candidate sets for the client to decrypt, refine, and re-upload.

More recently, homomorphic encryption has been integrated directly into encrypted database systems to compute over ciphertexts without continuous client intervention. For instance, Symmetria [30] utilizes multiplicative homomorphic encryption combined with additive extensions, whereas Rache [32] leverages layout-aware ciphertext placement alongside SIMD-optimized key switching techniques. Additional efforts from the applied cryptography community, such as HE3DB [5] and ArcEDB [40], introduce new arithmetic and logic operators to accelerate SQL evaluations over ciphertexts. However, these solutions often manifest as standalone programmatic scripts rather than being fully integrated into a standard relational database engine. Furthermore, to mitigate the extreme performance penalties and noise growth of FHE, recent query engines like NSHEDB [24] rely on word-level leveled homomorphic encryption. Despite these cryptographic advances, existing homomorphic databases typically operate on scalar records, encrypting individual database attributes into isolated ciphertexts. This scalar approach fails to exploit the parallel processing capabilities of modern FHE schemes, leading to substantial storage inflation and computational bottlenecks during query execution.

Hermes differs from prior systems by integrating homomorphic encryption directly into the ingestion path. Unlike approaches that rely on client-side ciphertext generation or immutable payload storage, Hermes exposes SQL-level UDFs for packing, encrypting, and mutating ciphertexts at runtime. Our system introduces a slot-aware data model that supports in-place ciphertext updates and maintains per-group local aggregates as part of the packing process.

2.2 Fully Homomorphic Encryption

Homomorphic encryption (HE) enables computation on encrypted data without decryption. The seminal work by Gentry [16] introduced the first fully homomorphic encryption (FHE) scheme, laying the foundation for later constructions such as BFV [8, 15], GSW [17], BGV [9], and CKKS [10]. TFHE [11] supports Boolean circuits and has seen recent improvements in bootstrapping performance [18, 33]. These schemes have been implemented in libraries including SEAL [31], HElib [21], OpenFHE [3], and LattiGo [26].

Hermes is built atop the BFV scheme [15], a lattice-based fully homomorphic encryption (FHE) construction supporting exact arithmetic over integers modulo a plaintext modulus t. Let $R = \mathbb{Z}[X]/(X^N + 1)$ be a cyclotomic ring of degree $N = 2^k$ for some $k \in \mathbb{N}$. A plaintext message is represented as an element of $R_t := R/tR$, while ciphertexts reside in R_q^2, where $q \gg t$ is a large ciphertext modulus.

A key feature of BFV is its support for *plaintext packing* via the Chinese Remainder Theorem (CRT). When $t \equiv 1 \mod 2N$, the

ring R_t splits as a direct product of $n = N/2$ slots:

$$R_t \cong \mathbb{Z}_t^n,$$

enabling SIMD-style evaluation of vectorized inputs. Each plaintext $\mathbf{m} = (m_0, \ldots, m_{n-1}) \in \mathbb{Z}_t^n$ is encoded into a single polynomial $m(X) \in R_t$, encrypted as $\mathsf{Enc}(m) = \mathbf{c} \in R_q^2$, and evaluated homomorphically across all slots.

Homomorphic operations preserve slot-wise semantics:

$$\mathsf{EvalAdd}(\mathsf{Enc}(\mathbf{m}), \mathsf{Enc}(\mathbf{m}')) = \mathsf{Enc}(\mathbf{m} + \mathbf{m}'),$$
$$\mathsf{EvalMult}(\mathsf{Enc}(\mathbf{m}), \mathsf{Enc}(\mathbf{m}')) = \mathsf{Enc}(\mathbf{m} \cdot \mathbf{m}'),$$

where the arithmetic is component-wise modulo t. In addition, the BFV scheme supports automorphism-based rotation operations, also known as *Galois rotations*, which cyclically shift slot positions:

$$\mathsf{EvalRotate}(\mathsf{Enc}(\mathbf{m}), r) = \mathsf{Enc}((m_{(i+r) \bmod n})_{i=0}^{n-1}).$$

Hermes leverages this packing structure to encode multiple database tuples into a single ciphertext. This design enables parallel evaluation, slot-wise updates, and sublinear aggregation. Importantly, all packing and update logic remains entirely within the encrypted domain, without leaking intermediate values or structural metadata.

In addition, the BFV scheme supports automorphism-based slot rotation using Galois group actions. For any $r \in \mathbb{Z}$, a cyclic slot rotation is realized via the Galois automorphism $\sigma_r : X \mapsto X^r \bmod (X^N+1)$, which acts on ciphertexts by permuting their internal slot layout. To evaluate this transformation homomorphically, the client must precompute and upload the corresponding Galois keys:

$$\mathsf{EvalAtIndex}(\mathsf{Enc}(\mathbf{m}), r) = \mathsf{Enc}((m_{(i+r) \bmod n})_{i=0}^{n-1}),$$

where the rotation index r corresponds to a specific automorphism in the Galois group $\mathrm{Gal}(R_q/\mathbb{Q})$. These keys enable Hermes to perform selective slot access, masking, and reorganization within encrypted vectors while preserving semantic security.

2.3 Alternative Secure Processing Paradigms

Alongside purely cryptographic storage, trusted hardware enclaves provide an alternative route for secure data processing. Systems utilizing Intel SGX such as ObliDB [14] and CryptSQLite [35] protect data in memory while offering oblivious query processing to prevent access pattern leaks. Enterprise solutions have also adopted this paradigm. For instance, recent iterations of Azure SQL Database [2] leverage enclaves to enable rich pattern matching over encrypted columns. Furthermore, hardware isolation techniques have been proposed to secure peripheral interfaces and web environments against side channel attacks [13]. While enclaves deliver high performance, they introduce strict hardware dependencies and remain vulnerable to sophisticated microarchitectural exploits, which pure homomorphic encryption avoids.

Another prominent paradigm involves Secure Multi-party Computation and federated privacy. Frameworks like Conclave [34] and Shrinkwrap [4] accelerate analytics over big data by partitioning queries between cleartext local processing and secure protocols. Similarly, advanced join aggregate queries can be evaluated across private datasets without exposing individual records [36]. Managing cross cryptographic leakages in such distributed databases is a complex endeavor [39]. Additionally, massive parallel query optimization strategies [37] and Private Information Retrieval [38] remain essential for scalable data extraction. However, multi-party protocols inherently require heavy network synchronization among non-colluding servers, which is a fairly strong assumption that may or may not hold for many practical scenarios. Hermes bypasses these communication bottlenecks by shifting the computational burden entirely to the server side evaluation of SIMD packed ciphertexts.

2.4 Terminology and Notation

We summarize key terms and notation used throughout this paper. Let n denote the number of slots supported by the plaintext modulus and ring parameters of the BFV encryption scheme. All vectors are implicitly assumed to be over $\mathbb{Z}_t^n$, where t is the plaintext modulus.

- **Slot:** Each ciphertext encodes a vector of n plaintext elements, known as *slots*. Slots are independently addressable under SIMD operations supported by BFV.
- **Group ID:** In Hermes, input tuples are partitioned by a logical *group ID*, which assigns each record to a packed vector instance. All records within the same group are encoded into the same ciphertext. The mapping from record to group is deterministic and index-driven.
- **Auxiliary Sum Slot:** Hermes reserves the final slot (slot $n-1$) in each ciphertext to store the *local sum*, i.e., the plaintext sum $\sum_{j=0}^{n-2} m_j$ of all payload slots. This value is computed at packing time and used to accelerate encrypted aggregation without Galois rotation.
- **Packed Representation:** A ciphertext that stores $n-1$ application values and one auxiliary statistic is referred to as a *packed representation*. This format enables efficient updates and aggregation via SIMD-aware primitives.
- **Logical Length:** Some plaintext vectors do not occupy all n slots. The number of meaningful entries is called the *logical length*, and is used to control serialization, summation, and slot masking operations.

3 Hermes: FHE Layer for Vectorized Tuples

Hermes is designed as a MySQL-integrated system with a new data abstraction for storing, updating, and querying encrypted relational data via fully homomorphic encryption (FHE). It is built upon the multi-slot property of lattice-based FHE schemes (e.g., BFV), where a single ciphertext can encode a vector of plaintexts. This allows Hermes to pack multiple database records into a single ciphertext, enabling efficient SIMD-style operations. To support dynamic updates, we develop homomorphic algorithms for inserting and deleting records within packed ciphertexts without decryption. Additionally, we maintain auxiliary local sums within the ciphertext layout to accelerate aggregate queries. The following sections detail the design of Hermes's packed ciphertext data model and the homomorphic primitives for mutability.

3.1 Packed Ciphertext Data Model

We propose a new packing-based layout where each ciphertext holds a fixed-size group of database records. This model amortizes encryption cost and supports efficient SIMD-style processing. To enable insert and delete operations over individual slots, we design

a homomorphic slot manipulation interface that combines masking, shifting, and slot-zeroing operations, all within the ciphertext algebra. Unlike conventional systems that require decryption and re-encryption for each update, our design ensures that updates can be applied entirely in the encrypted domain.

3.1.1 Slot-Aware Packing Strategy. In Hermes, we introduce a structured packing layout that enables efficient storage and manipulation of encrypted data within the SIMD-style slot architecture of the BFV homomorphic encryption scheme. Rather than encrypting each tuple into a separate ciphertext, which incurs significant memory and performance overhead, we adopt a batched layout where each ciphertext encodes a fixed-size vector of plaintext values, referred to as *slots*.

Let N denote the ring dimension of the BFV scheme. When batching is enabled (via the Chinese Remainder Theorem on cyclotomic rings), the scheme yields $n = N/2$ usable slots. For example, with $N = 8192$, Hermes can store $n = 4096$ individual scalar values per ciphertext. However, instead of utilizing all n slots for raw data, we reserve one designated slot (typically the last) to store an auxiliary value, such as the *local sum* of the vector entries. This yields an effective capacity of $n - 1$ logical records per ciphertext.

Each slot in a packed ciphertext has a fixed positional meaning: slot i corresponds to the i-th logical record in the group. This deterministic alignment enables precise slot-wise reasoning and efficient update semantics. When a ciphertext is decrypted, the unpacked vector reveals values at well-defined offsets, allowing Hermes to reconstruct the plaintext table fragment without additional metadata. This position-indexed design is critical for supporting secure in-place modifications, such as insertions and deletions, without leaking tuple identity or requiring explicit key-column mapping.

Formally, a packed ciphertext c in Hermes can be viewed as a vector:

$$c = \mathrm{Enc}([v_0, v_1, \ldots, v_{n-2}, \sigma]),$$

where v_i is the plaintext value in logical slot i, and σ is the slot reserved for auxiliary information, such as $\sigma = \sum_{i=0}^{n-2} v_i$ in the case of sum queries.

Compared to traditional row-wise encryption, this layout significantly improves storage density and amortizes encryption cost across multiple tuples. It also enables efficient homomorphic operations such as component-wise masking, conditional updates, and encrypted aggregation over slots. In essence, we reinterpret ciphertexts as *relational vectors* whose entries correspond to logical rows in a database group, thereby enabling vectorized query execution over encrypted domains.

3.1.2 Homomorphic Insertion. Hermes supports in-place insertion of plaintext values into packed ciphertexts without requiring decryption or client-side re-encryption. The goal is to insert a new value v into logical slot i of an existing ciphertext c, where slots i through $n - 2$ may already contain data. Since ciphertexts are encoded as vectors of fixed size, insertion requires shifting all subsequent entries rightward by one position to make room for the new value. This is analogous to an in-place insert into an array but performed entirely under encryption.

The insertion procedure consists of three homomorphic operations: (1) Galois rotation, (2) plaintext masking, and (3) ciphertext addition. We first perform a Galois right-rotation on c, yielding a new ciphertext c_{rot} in which all slots are shifted by one position. However, this operation naively shifts all slots, including the local sum at slot $n - 1$, which must be preserved. Therefore, Hermes applies a plaintext mask m_{preserve} to c before rotation to ensure that only slots i through $n - 2$ are moved. This yields a masked-and-rotated ciphertext c_{shifted} in which slot i is now vacant.

Next, the new plaintext value v is embedded into a one-hot plaintext vector m_v where $m_v[i] = v$ and $m_v[j] = 0$ for $j \neq i$. This mask is encrypted into c_v, a ciphertext containing v only at the desired slot. Finally, we compute:

$$c' = \mathrm{EvalAdd}(c_{\mathrm{shifted}}, \mathrm{EvalMult}(c_v, m_i)),$$

where m_i is a binary plaintext mask that ensures v is added only at slot i. This avoids interference with other slots.

In addition to updating the main data slots, Hermes updates the local sum σ stored at slot $n - 1$. This is achieved via scalar plaintext addition:

$$\sigma' = \mathrm{EvalAddPlain}(\sigma, v).$$

Since v is known in plaintext, the update to the auxiliary slot incurs negligible overhead and does not require re-encryption.

It should be noted that this procedure is performed within the database server process using OpenFHE's C++ API, without leaking which slot was modified or what value was inserted. The result is a new ciphertext that logically contains all previous values, v inserted at slot i, and an updated sum at slot $n - 1$. This mechanism allows Hermes to support encrypted ingestion of streaming data with minimal cryptographic cost.

When a ciphertext pack reaches the maximum number of supported slots, no further insertions can occur without creating a new ciphertext. In Hermes, we adopt a conservative policy: each group is associated with a single ciphertext, and insertion beyond capacity is rejected. Future extensions may support *chained packing*, where overflow records spill into a second ciphertext. However, doing so requires maintaining slot alignment metadata and reintroduces some of the per-ciphertext management overhead that Hermes aims to avoid.

3.1.3 Homomorphic Deletion. Hermes enables encrypted deletion over packed ciphertexts by simulating a logical left-shift over the underlying slots while preserving the ciphertext algebra. The goal is to remove the value at slot i of ciphertext c, and to maintain a compacted layout where all subsequent values are shifted leftward and the last slot is cleared. Importantly, this is done entirely within the encrypted domain, without re-encoding the ciphertext or exposing the deletion position to the server.

The deletion procedure consists of three conceptual steps: (1) rotation, (2) masking, and (3) auxiliary slot update. First, Hermes isolates the subvector of ciphertext c corresponding to slots $i + 1$ through $n - 2$ and rotates it leftward by one position. This is accomplished via a Galois left rotation:

$$c_{\mathrm{rot}} = \mathrm{EvalRotate}(c, -1),$$

followed by a masking step to ensure that only the intended region is affected. Specifically, a plaintext mask m_{shift} is applied before rotation to preserve slot integrity and avoid accidental overwriting of unaffected entries.

Next, Hermes applies a zeroing mask z to clear the final slot (slot $n-2$), which now contains a duplicate or stale value as a result of the shift. The mask z is a binary plaintext vector where $z[j] = 1$ for $j < n-2$ and $z[n-2] = 0$, ensuring:

$$c' = \text{EvalMult}(c_{\text{rot}}, z).$$

The slot at index i, which originally contained the value to delete, has now been overwritten by the shifted entry from $i+1$, and the final slot is set to zero, restoring the invariant that only $n-1$ data entries are active.

Finally, the local sum maintained in slot $n-1$ is updated by subtracting the deleted plaintext value v. Since v is assumed to be known at deletion time (e.g., from a query input), this update is executed via plaintext subtraction:

$$\sigma' = \text{EvalSubPlain}(\sigma, v).$$

All steps, including rotation, masking, and sum adjustment, are performed homomorphically within the database server using OpenFHE APIs. This design avoids decryption-reencryption cycles, ensures slot occupancy consistency, and maintains statistical privacy regarding which slot was deleted. As with insertion, the deletion logic treats ciphertexts as logical arrays and preserves relational semantics under homomorphic constraints.

When all payload slots in a packed ciphertext have been deleted, the ciphertext becomes logically empty, although its auxiliary sum slot may still contain residual values. In Hermes, such ciphertexts are treated as semantically inert: they contribute nothing to aggregate queries, and further deletions are treated as no-ops. However, the ciphertext itself is retained in the database to preserve structural consistency unless explicitly garbage-collected. To maintain correctness, Hermes ensures that the auxiliary sum is updated to zero whenever the payload becomes empty. This guarantees that global aggregation remains accurate even in the presence of entirely deleted groups.

3.2 Maintenance for Aggregate Queries

To support efficient encrypted aggregation, Hermes maintains a running *local sum* for each packed ciphertext. This design allows queries such as `SELECT SUM(attr)` or `GROUP BY key` to be evaluated via lightweight ciphertext addition without slot-wise traversal or Galois rotations. The remainder of this subsection describes how the local sum is constructed, maintained, and queried.

3.2.1 Encoding at Packing Time. Hermes adopts a hybrid encoding strategy in which each packed ciphertext stores not only a fixed-length array of plaintext values but also a precomputed auxiliary statistic: the sum of those values. This local sum is embedded into the last slot of the ciphertext at the time of packing, enabling efficient aggregation queries without the need for slot-wise traversal or ciphertext rotations (i.e., Galois automorphisms).

Given a batch of $n-1$ plaintext values $(v_0, v_1, \ldots, v_{n-2})$ to be packed into a ciphertext, Hermes computes their sum $s = \sum_{i=0}^{n-2} v_i$ in plaintext and constructs an extended vector of the form:

$$\mathbf{m} = (v_0, v_1, \ldots, v_{n-2}, s) \in \mathbb{Z}_t^n,$$

where t is the plaintext modulus and n is the total number of BFV batching slots (typically $n = N/2$ for a ring dimension N). The extended vector $\mathbf{m}$ is then passed to the OpenFHE API to generate a packed plaintext object $\text{Packed}(\mathbf{m})$, which is encrypted into a ciphertext $c = \text{Encrypt}(pk, \text{Packed}(\mathbf{m}))$.

This packing layout effectively reserves the final slot (slot $n-1$) as a local accumulator, denoted as $\text{sum}_{\text{local}}$. Because BFV ciphertexts support SIMD-style processing, all data and the auxiliary sum coexist in a single ciphertext. Since the sum is computed before encryption, it incurs no additional homomorphic cost.

The benefit of this encoding is twofold. First, it amortizes the cost of sum computation over all packed values, reducing the number of ciphertexts needed for downstream aggregation. Second, it enables highly optimized aggregate queries, such as `SELECT SUM(attr)`, by reducing global summation to a single ciphertext-level addition over the final slot of each ciphertext. Instead of rotating and adding up all slots, the query planner only needs to sum the values at slot $n-1$ across all ciphertexts:

$$\text{SUM(attr)} = \text{EvalAdd}(c_1[n-1], c_2[n-1], \ldots, c_k[n-1]).$$

This approach dramatically reduces computational complexity and latency, particularly when processing large encrypted datasets. By pre-allocating and embedding local statistics into the ciphertext layout, Hermes transforms aggregation from a slot-wise operation into a slot-agnostic one, leveraging the structure of FHE schemes for practical query acceleration.

3.2.2 Aggregation Update. While Hermes encodes a local sum at packing time, database workloads often require dynamic updates. To ensure the correctness of aggregation queries over mutable encrypted data, we design a lightweight incremental strategy to maintain the local sum as records are inserted or deleted. This strategy preserves consistency between the payload slots and the auxiliary sum slot, while avoiding costly recomputation or re-encryption.

Consider a ciphertext c whose first $n-1$ slots contain encrypted values $(v_0, \ldots, v_{n-2})$, and whose final slot (slot $n-1$) stores the corresponding local sum $s = \sum_{i=0}^{n-2} v_i$. When a new value v_{new} is inserted into the payload (e.g., at slot i), Hermes also adds v_{new} into the local sum via scalar plaintext addition. This is performed by constructing a plaintext vector $\mathbf{m}_{\text{aux}}$ such that:

$$\mathbf{m}_{\text{aux}} = (0, 0, \ldots, 0, v_{\text{new}}),$$

where only the final slot is nonzero. Then the updated ciphertext c' is computed homomorphically as:

$$c' = \text{EvalAdd}(c, \mathbf{m}_{\text{aux}}).$$

This operation increases the local sum while preserving the rest of the payload unchanged. Note that no re-encryption is required, and the update remains entirely within the ciphertext space.

A similar process applies to deletions. Suppose a value v_{del} is removed from a known slot. Hermes constructs a plaintext subtraction mask $\mathbf{m}'_{\text{aux}}$ where:

$$\mathbf{m}'_{\text{aux}} = (0, 0, \ldots, 0, -v_{\text{del}}),$$

and applies:

$$c' = \text{EvalAdd}(c, \mathbf{m}'_{\text{aux}}).$$

The net effect is that the auxiliary slot now holds the correct sum of the modified payload. Since inserted and deleted values are supplied in cleartext as part of the query expression, this approach avoids the overhead of decrypting and summing the full ciphertext.

This fine-grained sum adjustment strategy is a key enabler of real-time encrypted aggregation. It ensures that the auxiliary slot always remains synchronized with the payload, supporting fast 'GROUP BY' and 'SUM' queries without compromising correctness or requiring client interaction. All updates are executed inside SQL statements via homomorphic functions (e.g., MySQL UDFs) and do not reveal which slot was updated.

3.3 Security Analysis and Proof of Correctness

We provide a formal security analysis of Hermes under the standard IND-CPA model. Our design avoids client-server interaction after encryption and ensures that all slot-level operations are data-independent, thereby preventing leakage from slot position, access pattern, or update frequency. We further prove that the packed representation with embedded statistics does not weaken the semantic security of the base FHE scheme, and our masking/shift update primitives are indistinguishable under standard lattice assumptions.

3.3.1 Security Model and Assumptions. We analyze the security of Hermes under the standard indistinguishability under chosen-plaintext attack (IND-CPA) model, instantiated using BFV [15]. We assume a passive, honest-but-curious adversary with full access to encrypted database contents, including ciphertexts before and after updates, as well as query results. The adversary may observe ciphertext structure, slot layout, and numerical patterns in decrypted outputs but is assumed to follow protocol without actively deviating from the query logic. However, the server does not record low-level timing, memory access, or power traces: Preventing side-channel attacks is beyond the scope of this work.

In order to ensure that Hermes does not introduce any new leakage channels beyond what is inherent in the underlying FHE scheme, we require the following invariants to hold:

(1) **Indistinguishability of Ciphertexts:** For any two payload vectors v and v' of the same length, their packed ciphertexts $c = \texttt{Encrypt}(v)$ and $c' = \texttt{Encrypt}(v')$ are computationally indistinguishable under the IND-CPA security of BFV.
(2) **Update Pattern Privacy:** Insertion and deletion operations do not reveal the affected slot position or the identity of the modified value to the server. That is, for any two slots i and j, the adversary cannot distinguish an insertion at slot i from one at j by examining ciphertext differences alone.
(3) **Aggregation Consistency:** Global aggregation over auxiliary slots (e.g., encrypted 'SUM') reveals no more information than the final aggregate result; the per-ciphertext local sum values are protected under the encryption scheme.
(4) **No Adaptive Leakage:** Hermes does not reveal any intermediate plaintexts or randomness that could be used to link ciphertexts over time. All operations are performed homomorphically and reuse no client secrets post-encryption.

Importantly, the Hermes data model and update interface preserve these properties without relying on secure multi-party computation (MPC), secure enclaves, or trusted client-side computation. All operations, such as packing, insert/delete, and aggregation, are fully executed on encrypted data within the untrusted database server.

This model captures realistic cloud deployment scenarios where users outsource encrypted databases to an untrusted cloud provider but wish to retain strong confidentiality guarantees under standard cryptographic assumptions (e.g., Ring-LWE hardness). We note that this threat model is stronger than many property-preserving encryption (PPE) systems, which often leak order, frequency, or access pattern by design.

3.3.2 Preservation of IND-CPA Security. The semantic security of Hermes relies on the assumption that the underlying BFV scheme remains IND-CPA secure under all ciphertext manipulations performed in the system. This includes ciphertext packing, masked updates, homomorphic insertion and deletion, and the embedding of auxiliary local statistics such as the group sum. We now argue that Hermes does not introduce any structural leakage or functional dependencies that could be exploited to distinguish ciphertexts. The formal proof is based on the following two observations.

First, although Hermes embeds deterministic metadata (e.g., local sums) into specific slots, the entire ciphertext is encrypted using randomized BFV encryption over the full plaintext vector. IND-CPA security guarantees that the ciphertext remains indistinguishable even if some slots are derived from others, so long as the message vector is encrypted as a whole with fresh randomness. In particular, the inclusion of auxiliary statistics does not compromise the semantic security of any individual slot.

Second, Hermes performs all slot-level updates, such as insertion and deletion, purely through homomorphic operations. These updates use masking, shifting, and slot-wise additions, but never invoke decryption, re-encryption, or key switching. No new randomness is injected, and no plaintext values are revealed or reconstructed during runtime. As a result, each modified ciphertext maintains the same IND-CPA security level as its original version, with no additional leakage beyond the allowed ciphertext interface.

We start by recalling the notion of IND-CPA security.

Definition 3.1 (IND-CPA Security). Let $\mathsf{Enc}_{\mathsf{pk}}(m; r)$ denote the encryption of a plaintext m under public key pk and randomness r. A scheme is IND-CPA secure if no probabilistic polynomial-time (PPT) adversary $\mathcal{A}$ can distinguish between encryptions of m_0 and m_1 for any chosen pair of equal-length plaintexts, even given arbitrary access to the encryption oracle. Formally, for any PPT adversary $\mathcal{A}$, the advantage

$$\mathrm{Adv}_{\mathcal{A}}^{\text{IND-CPA}} = |\Pr[\mathcal{A}(\mathsf{Enc}(m_0)) = 1] - \Pr[\mathcal{A}(\mathsf{Enc}(m_1)) = 1]|$$

is negligible in the security parameter λ.

Lemma 3.2. *Let C be the set of ciphertexts generated and manipulated by Hermes. Then under the assumption that the base FHE scheme (BFV) is IND-CPA secure, the ciphertext $c \in C$ remains IND-CPA secure throughout all operations in Hermes.*

Proof. The proof proceeds by reduction. Suppose there exists an adversary $\mathcal{A}_{\text{Hermes}}$ that can distinguish two ciphertexts $c_0, c_1 \in C$ produced under Hermes' operations with non-negligible advantage. Then we construct an adversary $\mathcal{A}_{\text{BFV}}$ that breaks IND-CPA security of the underlying BFV scheme.

We consider the following components:

- **Packing**: When packing multiple values into a single ciphertext, we construct a vector $v = (v_0, \ldots, v_{n-2}, s)$ where $s = \sum_{i=0}^{n-2} v_i$. The encryption $\mathsf{Enc}(v)$ is indistinguishable from $\mathsf{Enc}(v')$ for any v' of same dimension, as this reduces to IND-CPA security over vector plaintexts in BFV.
- **Auxiliary Slot**: Including a local sum s in the last slot of the plaintext vector does not affect security, since s is deterministically computed from values already encrypted. Its inclusion does not leak new information beyond what is already present in the vector.
- **Insertion**: To insert a plaintext v into a packed ciphertext c, we encrypt v into a ciphertext c_v, construct a plaintext mask m with $m[i] = 1$, and compute $c' = c + \mathsf{EvalMult}(c_v, m)$. The resulting c' remains IND-CPA secure because it is the sum of IND-CPA secure ciphertexts and homomorphic evaluations thereof. The slot mask m is public and fixed, so no leakage is introduced.
- **Deletion**: Similar reasoning applies. The rotated ciphertext and masked-out tail are both outputs of deterministic homomorphic operations on encrypted data, preserving indistinguishability under the semantic security of BFV.
- **Aggregation**: Aggregation over auxiliary slots only involves ciphertext addition. Since BFV supports additive homomorphism, and all ciphertexts involved are IND-CPA secure, the sum remains indistinguishable.

Thus, any distinguishing advantage of $\mathcal{A}_{\text{Hermes}}$ implies an advantage in distinguishing $\mathsf{Enc}(v)$ vs $\mathsf{Enc}(v')$, contradicting the IND-CPA security of BFV. □

An important subtlety in Hermes is the deterministic embedding of auxiliary data into a fixed slot. While this creates a structural invariant, it does not violate IND-CPA since the data being embedded is functionally independent on other encrypted slots. From a cryptographic perspective, this corresponds to publishing a function $f(v_0, \ldots, v_{n-2}) = s$ along with the encryption of the vector, which is a standard practice in FHE-based secure computation. Furthermore, all masking and slot operations are purely homomorphic and do not involve conditional branching or secret-dependent control flow. No part of the computation leaks information via side channels, timing, or slot access patterns.

3.3.3 Update Privacy via Slot-Level Obfuscation. To preserve data obliviousness within packed ciphertexts, Hermes ensures that individual insertions and deletions are indistinguishable with respect to their slot index. Since encrypted updates are performed via uniform homomorphic operations, namely Galois rotations, plaintext masking, and slot-wise additions, the adversary cannot determine which slot was modified given a before-and-after pair of ciphertexts. We formalize this guarantee using a slot-level indistinguishability game.

We start by formally defining the game.

Definition 3.3 (Slot Update Indistinguishability Game). Let c be an encrypted packed ciphertext over n slots under BFV. Consider two update positions $i_0, i_1 \in \{0, \ldots, n-2\}$. The goal of the adversary $\mathcal{A}$ is to distinguish whether an insertion was applied at position i_0 or i_1. The game proceeds as follows:

(1) Challenger generates keypair $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^\lambda)$.
(2) Adversary chooses a plaintext vector $\vec{v} = (v_0, \ldots, v_{n-2})$ and computes $c \leftarrow \mathsf{Enc}_{\mathsf{pk}}(\vec{v})$.
(3) Challenger chooses a random bit $b \in \{0, 1\}$.
(4) Challenger inserts a known value δ at slot i_b using Hermes' homomorphic insert logic, resulting in $c' = \mathsf{Insert}(c, i_b, \delta)$.
(5) Adversary is given c' and outputs a guess $b' \in \{0, 1\}$.

We define the adversary's advantage as:

$$\mathrm{Adv}_{\mathcal{A}}^{\text{slot-ins}} = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$

Hermes guarantees slot-level update privacy if $\mathrm{Adv}_{\mathcal{A}}^{\text{slot-ins}}$ is negligible in the security parameter λ.

Theorem 3.4. *If the underlying BFV encryption scheme is IND-CPA secure, then Hermes satisfies slot-update indistinguishability for any polynomial-time adversary $\mathcal{A}$.*

Proof. We reduce the problem to the IND-CPA security of BFV and the semantic uniformity of the masking and shift primitives. Let m_i be a plaintext mask such that $m_i[j] = \delta$ if $j = i$, and 0 otherwise. The insertion operation computes:

$$c' = \mathsf{EvalAdd}(c_{\text{shifted}}, \mathsf{EvalMult}(c_\delta, m_i)),$$

where c_{shifted} is the result of a homomorphic rotation and masking on c that preserves all but slot i.

Because $c_\delta = \mathsf{Enc}(\delta)$ is fresh and semantically secure under BFV, and m_i is public and independent of δ, the term $\mathsf{EvalMult}(c_\delta, m_i)$ remains indistinguishable regardless of the value of i. The subsequent addition with c_{shifted} is also homomorphic and does not reveal slot index, since the masked slots are zeroed out beforehand. As a result, the output ciphertext c' is statistically indistinguishable whether insertion occurred at i_0 or i_1.

If $\mathcal{A}$ had non-negligible advantage in distinguishing i_0 from i_1, we could construct a distinguisher $\mathcal{B}$ that breaks IND-CPA security by embedding i_b into the encryption pattern. This contradicts our assumption. □

It is important to note that Hermes maintains *position-hiding homomorphism* not by permuting slot values, but by maintaining the slot structure while masking all auxiliary effects of an update. The Galois shift operator is position-invariant modulo a known offset; the zeroing mask ensures no unintentional data leaks from adjacent slots. The reserved auxiliary slot (e.g., local sum) is updated using scalar plaintext operations, and its growth reflects only the net value change, not the specific position of insertion or deletion. This makes Hermes suitable for applications requiring structural privacy in addition to semantic security.

3.3.4 Correctness of Encrypted Updates and Aggregates. We now formally prove that Hermes *correctly* maintains the semantics of encrypted vector updates and aggregate computations in the presence of insertions and deletions. Our objective is to show that:

(1) The logical payload vector remains consistent with intended insertions and deletions.
(2) The auxiliary sum slot reflects the current sum of all valid payload slots.
(3) The result of encrypted aggregation over multiple ciphertexts equals the true sum of the underlying plaintexts.

We assume a ciphertext $c \leftarrow \mathsf{Encrypt}([v_0, \ldots, v_{n-2}, \sigma])$, where $v_i \in \mathbb{Z}_t$ is the plaintext payload at slot i and $\sigma = \sum_{i=0}^{n-2} v_i$ is the precomputed local sum stored in the auxiliary slot $n-1$.

(1) Correctness of Homomorphic Insertion. Let $v_{\text{new}} \in \mathbb{Z}_t$ be a value inserted into position i (with $0 \le i \le n-2$). Define the updated plaintext vector as:

$$[v'_0, \ldots, v'_{n-2}] = [v_0, \ldots, v_{i-1}, v_{\text{new}}, v_i, \ldots, v_{n-3}]$$

The homomorphic insertion process rotates the suffix

$$\{v_i, \ldots, v_{n-2}\}$$

one position to the right using a Galois automorphism, masks the insertion point i with a one-hot vector $m^{(i)} \in \mathbb{Z}_t^n$ such that $m^{(i)}[i] = 1$ and 0 elsewhere, and adds $v_{\text{new}} \cdot m^{(i)}$ via plaintext multiplication. Let $c_{\text{ins}} = \mathsf{EvalAdd}(c_{\text{shifted}}, \mathsf{EvalMult}(\mathsf{Encrypt}(v_{\text{new}}), m^{(i)}))$. Then, under correct decryption:

$$\mathsf{Decrypt}(c_{\text{ins}})[j] = \begin{cases} v_j & \text{if } j < i \\ v_{\text{new}} & \text{if } j = i \\ v_{j-1} & \text{if } i < j \le n-2 \\ \sigma + v_{\text{new}} & \text{if } j = n-1 \end{cases},$$

which matches the intended semantics. Note that the local sum is homomorphically updated by

$$\sigma' = \sigma + v_{\text{new}} \in \mathbb{Z}_t,$$

through scalar plaintext addition on the last slot.

(2) Correctness of Homomorphic Deletion. Let v_i be deleted from position i. The deletion logic performs a Galois rotation that left-shifts all slots $j > i$, then applies a mask $z \in \mathbb{Z}_t^n$ such that $z[n-1] = 0$ and $z[j] = 1$ for $j < n-1$, to zero out the stale tail. The resulting ciphertext c_{del} decrypts to:

$$\mathsf{Decrypt}(c_{\text{del}})[j] = \begin{cases} v_j & \text{if } j < i \\ v_{j+1} & \text{if } i \le j \le n-3 \\ 0 & \text{if } j = n-2 \\ \sigma - v_i & \text{if } j = n-1 \end{cases},$$

where σ is again updated via scalar subtraction. The vector semantics remain preserved, and auxiliary statistics remain valid.

(3) Correctness of Global Aggregation. Let $c^{(1)}, \ldots, c^{(k)}$ be a collection of packed ciphertexts, each of form:

$$c^{(\ell)} = \mathsf{Encrypt}\left(\left[v_0^{(\ell)}, v_1^{(\ell)}, \ldots, v_{n-2}^{(\ell)}, \sigma^{(\ell)}\right]\right),$$

$$\sigma^{(\ell)} = \sum_{j=0}^{n-2} v_j^{(\ell)}.$$

Then, the encrypted global sum is computed by:

$$c_{\text{agg}} = \mathsf{EvalAdd}(c^{(1)}, \ldots, c^{(k)}),$$

followed by slot extraction $\mathsf{Extract}(c_{\text{agg}}, n-1)$ to isolate the final aggregated value.

The decrypted result satisfies:

$$\mathsf{Decrypt}(c_{\text{agg}})[n-1] = \sum_{\ell=1}^{k} \sigma^{(\ell)} = \sum_{\ell=1}^{k} \sum_{j=0}^{n-2} v_j^{(\ell)},$$

which matches the true plaintext sum across all tuples.

4 System Implementation

Dongfang: remove content that is duplicate of the previous section.

Dongfang: Update source code with head comments and update the lines of code in the following.

We implement Hermes as a suite of MySQL loadable functions written in C++ with OpenFHE v1.2.4. The system consists of modular components for slot-wise packing, homomorphic insertion and deletion, local-sum-aware aggregation, and slot-selective decryption, all of which are executed natively within the MySQL runtime.

As of March 2026, the codebase comprises 3,518 lines of code: including 2,045 lines of C++ core logic and the remainder for various scripts (e.g., Bash, Python, CMake). All components are implemented in C++ and compiled into dynamic libraries registered as MySQL UDFs. Our minimalist integration strategy avoids any modification to the MySQL source code or query engine internals. The full source code is publicly available on Github. Experiments and development were partially supported by the NSF-funded Chameleon Cloud infrastructure [25].

4.1 Architecture and Integration

Hermes is implemented as a runtime plugin using the MySQL User-Defined Function (UDF) interface. Each cryptographic operation is compiled as a C++ function into a shared object `.so` file, which is dynamically linked into the MySQL server process at runtime. This mechanism preserves full compatibility with unmodified MySQL 8.x deployments, requiring no server-side modifications or query proxies.

Each plugin function adheres to the MySQL UDF lifecycle. The `_init` function validates argument count and type, and initializes internal state if necessary. The core logic resides in `_func`, which executes per-row during the query evaluation. The optional `_deinit` function releases any memory or handles. Hermes uses this interface to expose cryptographic primitives including ciphertext packing, secure insertion and deletion, encrypted aggregation, and packed decryption.

In order to ensure compatibility between MySQL data types and OpenFHE plaintexts, all inputs are explicitly parsed within the plugin. MySQL UDF arguments arrive as raw C pointers and type tags, for example, `INT_RESULT` and `STRING_RESULT`. Hermes implements strict parsing logic to convert these inputs into 64-bit signed integers, using `std::istringstream` when necessary. Each scalar value $v \in \mathbb{Z}$ is then embedded into a plaintext vector $\mathbf{m} \in \mathbb{Z}_t^n$, using OpenFHE's `MakePackedPlaintext` API with a fixed slot layout.

Hermes adopts a ring dimension N, yielding $n = N/2$ usable slots in the BFV batching scheme. One slot (typically slot $n-1$) is reserved for auxiliary metadata such as the local sum. All remaining $n-1$ slots are used to store plaintext values. This layout is encoded during packing and preserved during ciphertext mutation. Plugin-side logic ensures that no slot is overwritten inadvertently, and runtime bounds checking is enforced.

The encryption and decryption routines are invoked entirely within the MySQL server process. Ciphertext objects are stored in

memory for the duration of the query and are not serialized unless explicitly requested. For diagnostic purposes, Hermes returns string-encoded ciphertext metadata of the form:

```
"0xADDR (v, size=s)",
```

where ADDR is the memory address of the ciphertext, $v \in \mathbb{Z}$ is the decrypted payload, and s is the object size in bytes. This format is used for interactive debugging and was critical during early plugin validation.

To integrate OpenFHE v1.2.4, Hermes statically compiles the BFV module and its dependencies into a relocatable shared object. At runtime, shared libraries such as `libOPENFHEcore.so` are copied into MySQL's plugin directory and made visible to `mysqld` via an injected `LD_LIBRARY_PATH` set in a `systemd` override file. This guarantees correct dynamic linking on Ubuntu 24.04 LTS and avoids linker errors from unresolved symbols.

All plugin code is written in C++17, built via CMake with full support for debug symbols and module-wise compilation. To comply with MySQL plugin constraints, all external symbols are wrapped in `extern "C"` linkage and use plain pointer-based memory semantics. No exceptions are allowed across the FFI boundary. Hermes writes all diagnostic logs to `stderr`, which are forwarded into the MySQL error log at `/var/log/mysql/error.log` for offline inspection.

4.2 Context and Key Management

Hermes relies on the BFV scheme as implemented in OpenFHE v1.2.4. To ensure correct and consistent behavior across all plugin functions, we construct a single shared encryption context that is initialized at plugin load time and reused throughout the query lifecycle. This context is created using OpenFHE's `GenCryptoContext` API, parameterized with a plaintext modulus t, ring dimension N, and multiplicative depth d. In our implementation, we set $t = 65537$, $N = 2^{14}$, and $d = 2$, which balances noise growth and packing efficiency.

The encryption context C defines the underlying polynomial ring $R = \mathbb{Z}_q[x]/(x^N + 1)$, as well as the modulus chain $q = q_1 \cdot q_2 \cdots q_L$ used for leveled homomorphic operations. Once C is created, we enable the following scheme features:

- Public key encryption (PKE) for basic encryption and decryption;
- Leveled SHE for both plaintext-ciphertext and ciphertext-ciphertext arithmetic;
- Advanced SHE for evaluation keys including Galois and relinearization keys.

Key generation is performed once at startup using `KeyGen()`, yielding a public key pk and a secret key sk. We also generate auxiliary keys:

- Relinearization keys `EvalMultKeyGen`(sk) for ciphertext size reduction;
- Rotation keys `EvalSumKeyGen`(sk) and `EvalAtIndex`(sk, i) for slot manipulation.

To prevent accidental key regeneration or state corruption across plugin invocations, all keys and contexts are encapsulated in static C++ global variables guarded by initialization flags. This ensures that the context is instantiated only once, and reused by all subsequent invocations of Hermes UDFs. All keys are stored in memory and are not serialized to disk, thereby avoiding potential I/O or deserialization bugs during runtime.

In addition to functional keys, Hermes supports a default plaintext packing policy where a vector $\mathbf{v} = (v_0, v_1, \ldots, v_{n-2})$ is padded with an auxiliary local sum $v_{n-1} = \sum_{i=0}^{n-2} v_i$. This logic is executed at plaintext construction time and preserved across all homomorphic operations.

4.3 Ciphertext Format and Memory Layout

Hermes adopts a packing model based on the batching technique available in BFV, which maps plaintext vectors into the ring $R_t = \mathbb{Z}_t[x]/(x^N + 1)$ via the Chinese Remainder Theorem (CRT). For a ring dimension N, this results in $n = N/2$ usable plaintext slots, each of which can store an individual integer mod t.

Each ciphertext in Hermes is treated as a structured container with fixed semantic fields:

- Slots $[0, \ldots, n-2]$: store plaintext database records (e.g., salary values).
- Slot $n-1$: reserved for auxiliary metadata (e.g., local sum).

This layout enforces slot-position invariants critical for query semantics. For example, slot i always corresponds to the i-th tuple in a logical group. When a slot is modified, its relative position remains unchanged; thus, operations such as `INSERT` and `DELETE` preserve the integrity of the ciphertext as a vector block.

To ensure predictable memory usage and interoperation with MySQL, each ciphertext object is allocated on the heap and remains opaque to MySQL. Rather than serializing ciphertexts into strings or blobs, Hermes returns a debug string containing:

- The memory address of the ciphertext (as a hexadecimal pointer).
- A plaintext verification value (via local decryption).
- The ciphertext object size (from C++ `sizeof` operator).

This enables validation and tracing without performing I/O. For example, an encrypted salary vector might yield the following result:

```
0x7ffee31b2ac0 (sum = 13200, size = 128).
```

Internally, all ciphertexts are represented using OpenFHE's polymorphic type system. The type `Ciphertext<DCRTPoly>` encapsulates a leveled ciphertext vector over RNS-encoded polynomial rings. While OpenFHE supports deep serialization, we avoid this in Hermes to minimize dependency on runtime I/O.

Slot-level indexing is consistent across all plugin operations. For example:

- `HERMES_PACK_CONVERT()` maps a plaintext array into slots $[0, \ldots, n-2]$ and computes a local sum into slot $n-1$.
- `HERMES_INSERT()` and `HERMES_REMOVE()` manipulate only targeted slots while preserving untouched regions.
- `HERMES_SUM()` reads only slot $n-1$ across ciphertexts and aggregates via `EvalAdd`.

By adopting this aligned memory and slot layout, Hermes enables relational semantics over encrypted vectors with minimal runtime overhead. Slot position serves as both a logical index and a security-invariant boundary, allowing homomorphic updates to be computed without decoding or leaking positional access.

4.4 Encrypted Aggregate Query

Hermes supports encrypted evaluation of SQL-style aggregate queries by leveraging the auxiliary local sum slot embedded in each packed ciphertext. This slot, typically at index $n-1$, stores the sum of all payload values within the ciphertext. As described in earlier sections, this local sum is precomputed during packing and maintained incrementally across insertions and deletions. The aggregation engine relies on this invariant to compute full-table or group-wise sums without rotating or scanning individual slots.

In order to compute a global sum over a table, Hermes executes ciphertext-level additions using OpenFHE's `EvalAdd` API. Given a collection of ciphertexts $\{c^{(1)}, c^{(2)}, \ldots, c^{(k)}\}$, each with its local sum $\sigma^{(\ell)}$ stored in slot $n-1$, the global sum is computed as:

$$\sigma_{\text{global}} = \text{Decrypt}\left(\left[\texttt{EvalAdd}(c^{(1)}, \ldots, c^{(k)})\right][n-1]\right).$$

This implementation is exposed to MySQL via the UDF of global sum `HERMES_PACK_GLOBAL_SUM()`. The function accepts a sequence of ciphertext pointers corresponding to groups (or full-table partitions), adds them homomorphically, and returns the resulting ciphertext encoding the total sum in the last slot. The decryption step is performed separately using `HERMES_DEC_VECTOR_SUMMABLE()`, which extracts the $n-1$-th slot and converts it into a SQL-visible scalar.

For group-wise aggregation (e.g., `GROUP BY department`), Hermes relies on a grouping UDF `HERMES_PACK_GROUP_SUM()`, which performs group-level ciphertext addition using MySQL's internal grouping operator. In this mode, each group maintains its own packed ciphertexts with preserved slot structure. The UDF dispatch logic ensures that ciphertexts with the same group key are aggregated homomorphically, producing per-group sum ciphertexts with slot $n-1$ holding the encrypted result.

All aggregation logic is rotation-free. Unlike standard FHE aggregation strategies that require $\log n$ Galois rotations and relinearizations to sum all slots within each ciphertext, Hermes uses only one ciphertext addition per group and one decryption at the end. This leads to significant runtime savings, particularly when $k \gg n$ and group sizes are unbalanced.

Internally, Hermes maintains a fixed slot indexing contract: only slot $n-1$ may be accessed for aggregation purposes. This enforces consistent semantics and allows plugin logic to treat ciphertexts as opaque blobs outside the final slot. As a result, Hermes's aggregation functions are modular, efficient, and fully SQL-compatible under encrypted execution.

4.5 Homomorphic Update

Hermes supports in-place updates to packed ciphertexts by implementing homomorphic analogues of insertion and deletion. These operations allow the system to modify encrypted vectors directly within the database engine, without requiring decryption or client-side recomputation. The core principle is to simulate array-like slot operations, such as shifting, overwriting, and nulling, entirely within the ciphertext algebra provided by the BFV scheme.

Insertion is modeled as a three-step procedure. First, the tail of the vector (from the insertion index i to slot $n-2$) is rotated one position to the right using a Galois automorphism. Second, a new value is encrypted into a single-slot vector and masked into position i using a one-hot plaintext mask. Third, the auxiliary slot is incremented homomorphically by the inserted value, using scalar plaintext addition on slot $n-1$. All components are assembled using OpenFHE's native operators: `EvalRotate`, `EvalMult` with plaintext masks, and `EvalAddPlain`.

Deletion follows the inverse pattern. The vector suffix

$$[i+1, \ldots, n-2]$$

is rotated left by one slot, overwriting the deleted position. A plaintext mask then zeros out the final payload slot to maintain the invariant that only $n-1$ slots are active. The local sum slot is adjusted via plaintext subtraction. The deleted value must be known in plaintext at deletion time to update the sum correctly, which is compatible with SQL's explicit deletion semantics (e.g., `WHERE attr = v`).

Both insertion and deletion operations are stateless with respect to the plugin logic. The update functions accept a ciphertext pointer, a plaintext value, and a slot index as input. All intermediate ciphertexts, such as rotated and masked versions, are constructed on the fly and discarded after use. This minimizes memory footprint and avoids cross-query state retention.

The update engine enforces strict bounds checking and slot alignment. No insertion is allowed at slot $n-1$, which is reserved for auxiliary metadata. All index operations are validated against the configured slot count during context initialization. This prevents buffer overflows or malformed ciphertexts due to user error or adversarial input.

Unlike vectorized plaintext arrays, encrypted slot operations must preserve both algebraic structure and semantic security. Hermes ensures this by reusing known-safe operations from the BFV scheme and by avoiding conditional branching based on encrypted values. All update paths are data-oblivious at the slot level: the pattern of rotations and additions depends only on public slot indices and user-provided plaintexts, never on encrypted contents.

4.6 Decryption Interface and Result Extraction

Hermes performs decryption entirely within the MySQL server process, using the BFV secret key generated at plugin initialization. Since all ciphertexts follow a structured packing layout, the decryption interface focuses on slot-specific result recovery, rather than full plaintext vector traversal. The most common operation is extracting the auxiliary slot $n-1$, which holds the local or aggregated sum over encrypted database values.

The decryption workflow begins with the invocation of a dedicated decryption function, which receives a ciphertext handle as input. The handle is passed as a hexadecimal string from SQL, identifying an in-memory ciphertext stored in the plugin's internal heap. After verifying the validity of the pointer and its alignment with the expected OpenFHE type system, the function decrypts the ciphertext using the user's secret key.

Decryption is implemented using OpenFHE's `Decrypt` API, followed by conversion into a plaintext vector using `SetLength` and `GetPackedValue`. Once the plaintext vector is recovered, only the desired slot is extracted, usually slot $n-1$ for aggregate queries or slot i for targeted inspection. The result is then formatted as a scalar value and returned to MySQL via the UDF return buffer.

All decryption paths are read-only and do not mutate the ciphertext or context state. This ensures that repeated invocations yield identical results and remain free of side effects. To enforce security guarantee, Hermes never exposes the full plaintext vector to the SQL interface; only a single scalar per call is returned. This protects against mass leakage through oversized responses or implicit slot traversal.

The plugin logic also includes bounds checking on the slot index. If a user attempts to extract from an invalid slot (e.g., outside $[0, n-1]$), the decryption function returns `NULL` and emits a warning in the MySQL error log. This design ensures fail-safe behavior in case of malformed input or version mismatch between packing and unpacking logic.

Hermes avoids automatic decryption of entire ciphertext arrays. For use cases requiring multiple slot values, the system recommends issuing repeated scalar decryptions rather than exposing full vector access. This restriction enforces minimal data release and is consistent with the threat model that permits aggregate leakage but not tuple-level reconstruction.

4.7 Challenges and Leaerned Lessons

MySQL's user-defined function (UDF) interface, while technically C-compatible, imposes a range of low-level constraints that complicate integration with modern C++ libraries like OpenFHE. One prominent example is the UDF return buffer: although MySQL allows `char*` return types, it silently allocates a default buffer of 256 bytes and expects the result to be manually null-terminated and sized via the `length` pointer. If we forget to carefully track the length or overflows the buffer, the output is truncated or causes undefined behavior, with no warning from MySQL. To ensure safe operation, Hermes explicitly monitors string lengths and uses `memcpy` with zero padding.

Another challenge arises from MySQL's strict type expectations during UDF registration and invocation. The `arg_type[]` array must precisely match the declared input behavior, and any mismatch, e.g., expecting `INT_RESULT` but receiving a string-cast input, can result in either incorrect parsing or segmentation faults. This sensitivity is further exacerbated when dealing with numeric values passed as `STRING_RESULT`, requiring explicit string-to-integer conversions inside the plugin. Hermes handles this by explicitly checking and parsing each argument type and issuing warnings for unsupported patterns.

Despite being written in C++, all Hermes UDFs must be registered using `extern "C"` linkage, preventing name mangling and ensuring compatibility with MySQL's dynamic symbol resolution. However, this also restricts the use of modern C++ idioms such as RAII, smart pointers, or class encapsulation for memory management. Instead, plugin state (e.g., ciphertext handles, encryption context) must be tracked manually via global heap allocations or pointer-passing, with careful attention to lifecycle management to avoid memory leaks or dangling references across UDF calls.

Linking OpenFHE into the MySQL runtime poses its own set of challenges. Since MySQL does not inherit environment variables like `LD_LIBRARY_PATH` when started as a systemd service, shared libraries such as `libOPENFHEpke.so` are not found unless explicitly injected. Hermes addresses this by generating a `systemd` override file to add the plugin directory to the runtime path and restarting the MySQL daemon after deployment. This step is essential but poorly documented, and its omission results in non-descriptive runtime errors about unresolved symbols in the shared object.

A particularly subtle issue involves the generation and use of Galois keys for slot rotation operations. OpenFHE requires that Galois keys be generated explicitly after context and secret key initialization, and failure to do so results in undocumented runtime errors or silent failures during rotation-based operations. Moreover, if multiple plugins generate conflicting Galois keys or overwrite global state, the system may enter a corrupted state, especially if MySQL reuses shared library handles across multiple function invocations. Hermes mitigates this by generating all required keys exactly once at system initialization and disallowing repeated or partial regeneration during runtime.

Table 1: Summary of evaluation datasets

Dataset	Domain	#Tuples	Value Range
COVID-19 [12]	Healthcare	341	~[0, 100k]
hg38 [22]	Genomics	34,424	~[0, 10k]
Bitcoin [6]	Finance	1,086	~[0, 2^{31}]

5 Evaluation

5.1 Experimental Setup

All experiments are conducted on a Dell PowerEdge R6525 server provisioned via the CHI@UC Chameleon testbed [25]. The compute node is equipped with dual AMD EPYC 7763 processors operating at 2.45GHz, providing a total of 128 physical cores and 256 hardware threads, alongside 512GB of memory. The system runs on Ubuntu 24.04 LTS and utilizes MySQL version 8.0.42 as the underlying relational database engine. The homomorphic cryptographic operations are implemented using OpenFHE version 1.2.4 configured with the BFV scheme. For all evaluations, we establish the encryption context using a default ring dimension of $N = 2^{14}$ and a plaintext modulus of $t = 2^{16}$, which yields exactly $n = 8192$ plaintext slots per ciphertext for vectorized execution.

We evaluate Hermes on three real-world datasets that represent diverse application domains and data distributions, summarized in Table 1. The *COVID-19* dataset contains 341 daily records of pandemic-related statistics in the U.S., such as hospitalizations and case counts. The *hg38* dataset corresponds to genomic annotations in the human reference genome, consisting of over 34,000 entries such as transcription start sites and exon counts. The *Bitcoin* dataset records cryptocurrency trade volumes on a 3-day interval from 2013 to 2022. Since raw Bitcoin trade values are extremely large, we apply a rescaling factor of 1/24 to obtain minute-level approximations before encryption. All datasets are stored as single-attribute tables in a MySQL database. Hermes operates over these tables using only SQL-based UDFs, without modifying schema definitions or client-side logic. We use a packing size of 8192 slots throughout all experiments, and all queries are executed on a single-node MySQL server with OpenFHE v1.2.4.

To comprehensively evaluate the architectural and algorithmic advantages of Hermes, we establish four distinct baselines. We

partition these baselines into end-to-end system macrobenchmarks executed within the database engine and isolated cryptographic microbenchmarks.

For the end-to-end evaluations within MySQL, we compare Hermes against a Singular FHE approach and a Plaintext baseline. The Singular FHE baseline represents a naive integration where each database tuple is encrypted into an independent ciphertext without SIMD batching. This configuration highlights the computational and storage overheads of unpaced encryption and justifies our vectorized data model. The Plaintext baseline operates on completely unencrypted data using standard SQL queries. It serves as the absolute performance ceiling to quantify the exact system-level penalty introduced by our cryptographic layer.

To isolate our algorithmic improvements from the inherent execution overhead of the database engine, we implement two additional baselines as standalone C++ microbenchmarks. The first is a Standard Packed FHE baseline, which utilizes the same multi-slot ciphertext layout as Hermes but relies on the conventional sequence of logarithmic Galois rotations and masking operations to compute aggregate sums. This comparison isolates the latency reduction achieved specifically by our auxiliary local sum embedding technique. The second microbenchmark employs the Paillier cryptosystem [27], representing the traditional Partially Homomorphic Encryption (PHE) standard for secure database aggregations. Because Paillier supports only scalar additive homomorphism, this evaluation demonstrates how the vectorized parallelism of modern FHE outpaces legacy scalar schemes in analytical throughput.

To guide the remainder of our evaluation, we structure our experiments around four core research questions (RQs):

- RQ1: How does the vectorized packing data model improve initial data ingestion and encryption throughput compared to singular tuple encryption? We address this in Section 5.2.
- RQ2: What is the computational latency of performing secure in-place modifications such as record insertions and deletions over packed ciphertexts? We explore this in Section 5.3.
- RQ3: How does the auxiliary local-sum embedding technique perform against traditional cryptographic aggregation queries in isolated microbenchmarks? We answer this in Section 5.4.
- RQ4: How does the logical group capacity influence the overall execution latency across different relational operators? We analyze this in Section 5.5.

5.2 Encryption Throughput

To answer RQ1, we evaluate the initial data ingestion performance by focusing strictly on the cryptographic overhead. We compare the Hermes vectorized packing model at its boundary capacities against the Singular FHE baseline. Figure 1 presents both the absolute time required to encrypt the complete tables and the normalized throughput measured in tuples per second. The vertical axes utilize a logarithmic scale to properly depict the massive performance gap.

The Singular FHE baseline forces the database to generate an independent ciphertext for every single tuple. As shown in Figure 1(c), encrypting the hg38 dataset tuple by tuple consumes over 4048 seconds, yielding a slow throughput of merely 8 tuples per second. By contrast, Hermes leverages the SIMD capabilities of the BFV scheme to pack multiple relational tuples into a single ciphertext. Even under the most conservative structural configuration of 128 slots, the encryption time for hg38 drops to 35 seconds, boosting the throughput to 958 tuples per second.

When operating at the optimal packing scale of 4096 slots, Hermes reduces the total encryption time for hg38 to 1.2 seconds. The overlaid throughput trajectory highlights this dramatic efficiency gain, revealing a high processing rate of 27783 tuples per second. Similar computational boundaries are observed in the Bitcoin and COVID-19 datasets. These measurements confirm that the vectorized data model successfully reduces the fundamental throughput bottleneck of applying homomorphic encryption to relational database systems.

5.3 In-Place Update Latency

To answer RQ2, we evaluate the system performance for in-place data modifications. We focus on the amortized latency per tuple to accurately reflect the throughput capabilities of the vectorized packing architecture under bulk operation workloads.

Figure 2 presents the amortized latency for insertion operations. The Singular FHE baseline exhibits a severe bottleneck, requiring approximately 124 milliseconds to insert a single tuple. This delay stems from the mathematical requirement to generate a complete polynomial ciphertext for every isolated record. Hermes leverages the SIMD slots to distribute this cryptographic overhead. Under the optimal configuration of Hermes-4096, the amortized insertion time is reduced to approximately 0.03 milliseconds per tuple. This structural advantage translates to an over 4000x speedup for the Bitcoin dataset and similar gains across the other workloads.

Figure 3 illustrates the performance of deletion operations. In the Singular FHE system, a deletion is a direct SQL drop operation requiring merely 4 milliseconds. Hermes cannot physically drop rows within a packed ciphertext and must instead execute homomorphic masking to erase specific slots. Although a single packed deletion takes around 54 milliseconds to compute, amortizing this operation across 4096 slots yields an effective latency of 0.013 milliseconds per tuple. As highlighted by the overlaid metrics, this results in a 369x speedup over the singular approach for the hg38 dataset. Other two datasets show similar trends, leading to speedups around 300x.

These empirical results validate that the SIMD packing design does not degrade the practical update performance. By amortizing the cryptographic polynomial operations across thousands of relational tuples, Hermes successfully maintains sub-millisecond effective latencies for both insertions and deletions.

5.4 Cryptographic Aggregation Queries

To answer RQ3, we evaluate the performance of Hermes on aggregation queries. To isolate the intrinsic cryptographic efficiency of our data model from the inherent execution overhead of the MySQL engine, we conduct these evaluations at the core operator level. Although Hermes natively processes SQL aggregations, measuring them end-to-end would introduce parsing and scheduling artifacts that obscure the pure cryptographic throughput. Therefore, we implement this section's benchmarks as standalone C++ programs.

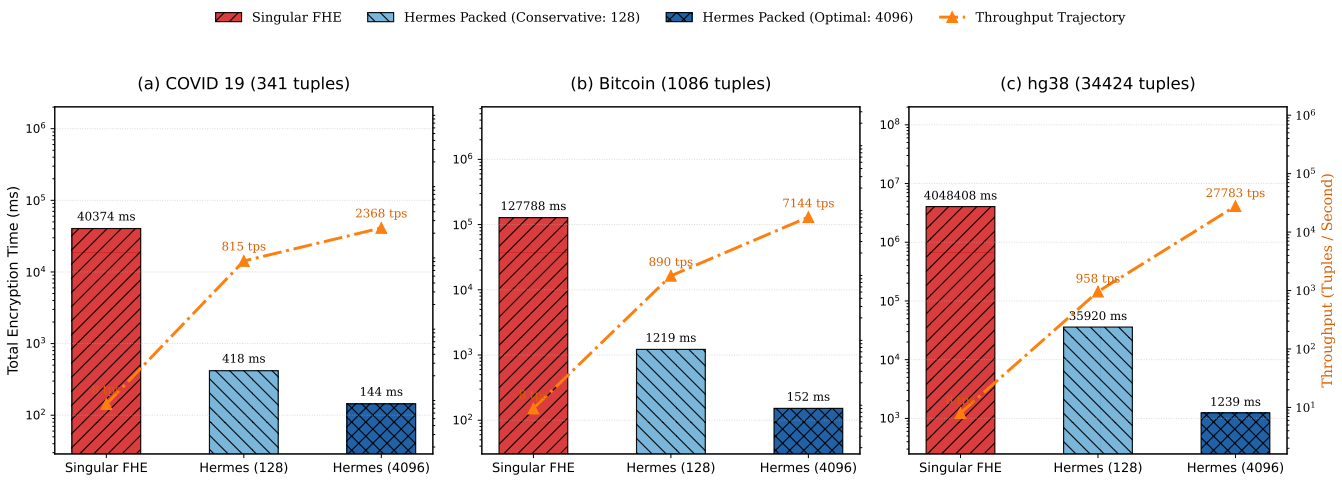

Figure 1: Encryption throughput and absolute time comparison across three datasets. Subfigures (a), (b), and (c) display the evaluation for the Singular FHE baseline, the conservative Hermes configuration at 128 slots, and the optimal Hermes configuration at 4096 slots. The left vertical axes and bars represent absolute encryption time in milliseconds. The right vertical axes and the overlaid line graphs track the processing throughput measured in tuples per second (tps). All vertical axes utilize a logarithmic scale.

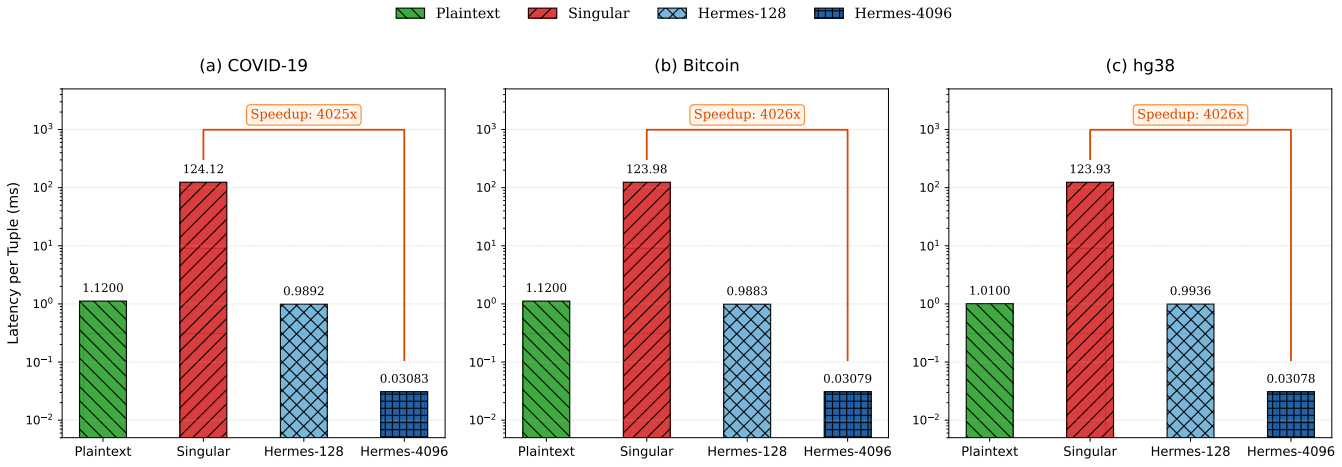

Figure 2: Amortized in-place insertion latency per tuple across three evaluation datasets. The vertical axes utilize a logarithmic scale. The floating labels denote the relative speedup of Hermes-4096 against the Singular FHE baseline.

This approach allows us to directly compare our rotation-free aggregation technique against traditional cryptographic methods under strict computational isolation. All benchmarks are executed with $N = 2^{14}$ and $t = 65537$, yielding 8192 slots.

Figure 4(a) compares the throughput of encrypted addition between the proposed Hermes data model and the Intel Paillier Cryptosystem Library (IPCL) [23]. While Paillier is a highly optimized partially homomorphic scheme, its scalar nature requires n separate ciphertext additions to process n tuples. As the packing scale increases to 8192 slots, Paillier's latency grows linearly to 120.5 ms. In contrast, Hermes utilizes the SIMD capabilities of BFV to perform a single vector-add operation that processes all 8192 slots simultaneously. This results in a near-constant latency of 1.57 ms, representing a 76x throughput advantage at the maximum scale.

Figure 4(b) evaluates the scalability of encrypted aggregation. Standard FHE aggregation strategies rely on a sequence of $\log n$ Galois rotations and relinearizations to sum internal slots. At 8192 slots, this conventional approach consumes over 1165 ms per ciphertext. Hermes fundamentally bypasses this bottleneck by embedding a precomputed local sum into a reserved auxiliary slot during the packing phase. Global aggregation in Hermes is thus reduced to a single ciphertext-level addition over these auxiliary slots. As shown in the results, Hermes completes the same aggregation in 1.58 ms, achieving a 738x speedup over the rotation-based baseline.

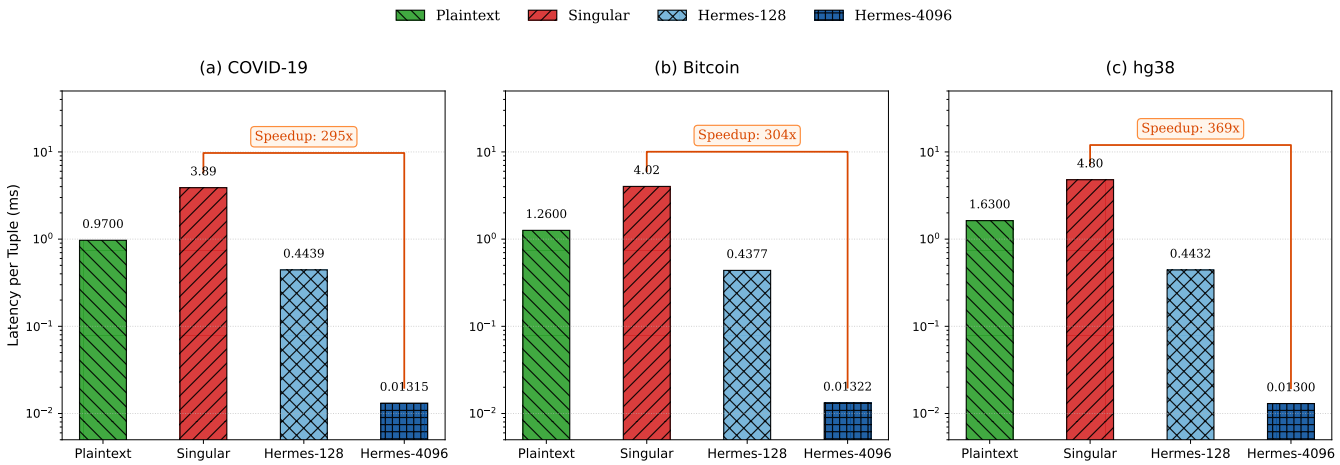

Figure 3: Amortized in-place deletion latency per tuple across three evaluation datasets. The vertical axes utilize a logarithmic scale. The floating labels denote the relative speedup of Hermes-4096 against the Singular FHE baseline. **Dongfang: The 'Speedup' line needs to be elevated: it's cutting the numbers.**

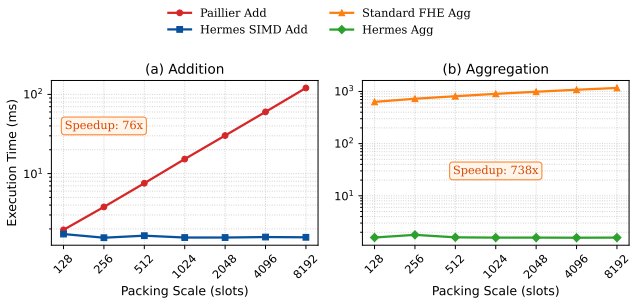

Figure 4: Cryptographic microbenchmarks comparing (a) vector-add throughput against Intel-accelerated Paillier and (b) Hermes rotation-free aggregation against the standard logarithmic-rotation FHE baseline. Both vertical axes use a logarithmic scale.

These microbenchmarks demonstrate that the performance gains in Hermes are not merely implementation-level optimizations but stem from a structural shift in how homomorphic primitives are applied to relational data. By aligning the database layout with the algebraic invariants of SIMD-FHE, Hermes transforms traditionally expensive cryptographic tasks into constant-time operations.

5.5 Scalability of Packing Granularity

To address RQ4, we analyze the system performance across different logical group capacities ranging from 128 to 4096 slots. Figure 5 illustrates the execution latency for encryption, insertion, and deletion operations under varying packing scales.

Figure 5(a) demonstrates a clear inverse relationship between the packing scale and the total encryption time. As the packing capacity doubles, the number of required physical ciphertexts is reduced by half. This leads to an exponential reduction in the overall cryptographic overhead. For example, encrypting the complete hg38 dataset takes over 35 seconds with a pack size of 128 but drops to approximately 1.2 seconds when the scale expands to 4096. This validates the design philosophy that maximizing the utilization of plaintext slots effectively amortizes the initial data ingestion latency.

Conversely, Figure 5(b) and Figure 5(c) reveal that the performance of in-place modifications remains entirely decoupled from the packing scale. The time required to perform 100 insertions or 100 deletions stays roughly constant across all scales. Specifically, the batch insertions stabilize at approximately 12.6 seconds, while the corresponding deletions take about 5.4 seconds. This constant time complexity is a direct manifestation of the underlying homomorphic mechanics. Because the polynomial ring dimension is fixed at $N = 2^{14}$ during the cryptographic context initialization, the computational cost of shifting and masking operations depends solely on the algebraic structure rather than the logical payload. Processing a ciphertext containing 128 active slots requires the exact same CPU cycles as processing one with 4096 active slots. Consequently, system administrators can safely maximize the packing scale to optimize storage footprints and encryption throughput without incurring any latency penalties during subsequent data modifications.

6 Conclusion and Future Work

Dongfang: TODO

References

[1] Adi Akavia, Neta Oren, Boaz Sapir, and Margarita Vald. 2023. CSHER: A System for Compact Storage with HE-Retrieval. In *Proceedings of the 32nd USENIX Conference on Security Symposium*. USENIX Association.

[2] Panagiotis Antonopoulos, Arvind Arasu, Kunal D. Singh, Ken Eguro, Nitish Gupta, Rajat Jain, Raghav Kaushik, Hanuma Kodavalla, Donald Kossmann, Nikolas Ogg, Ravi Ramamurthy, Jakub Szymaszek, Jeffrey Trimmer, Kapil Vaswani, Ramarathnam Venkatesan, and Mike Zwilling. 2020. Azure SQL Database Always Encrypted. In *Proceedings of the 2020 ACM SIGMOD International Conference on Management of Data* (Portland, OR, USA) *(SIGMOD '20)*. Association for Computing Machinery, New York, NY, USA, 1511–1525. doi:10.1145/3318464.3386141

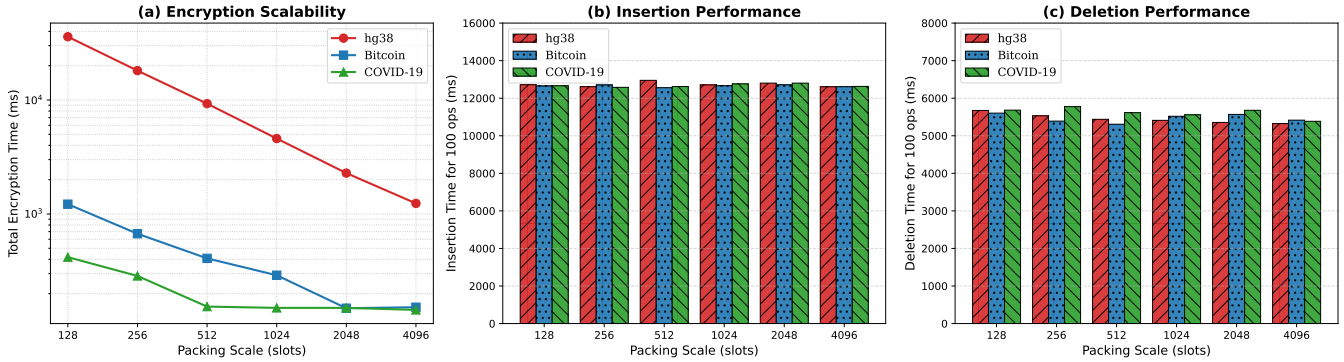

Figure 5: Performance scalability of Hermes across varying packing scales from 128 to 4096 slots. Subfigure (a) shows the total encryption time in a log-log scale. Subfigures (b) and (c) display the execution time for 100 in place insertions and deletions respectively.

[3] Ahmad Al Badawi, Jack Bates, Flavio Bergamaschi, David Bruce Cousins, Saroja Erabelli, Nicholas Genise, Shai Halevi, Hamish Hunt, Andrey Kim, Yongwoo Lee, Zeyu Liu, Daniele Micciancio, Ian Quah, Yuriy Polyakov, Saraswathy R.V., Kurt Rohloff, Jonathan Saylor, Dmitriy Suponitsky, Matthew Triplett, Vinod Vaikuntanathan, and Vincent Zucca. 2022. OpenFHE: Open-Source Fully Homomorphic Encryption Library. Cryptology ePrint Archive, Paper 2022/915. https://eprint.iacr.org/2022/915.

[4] Johes Bater, Xi He, William Ehrich, Ashwin Machanavajjhala, and Jennie Rogers. 2018. Shrinkwrap: efficient SQL query processing in differentially private data federations. *Proc. VLDB Endow.* 12, 3 (Nov. 2018), 307–320. doi:10.14778/3291264.3291274

[5] Song Bian, Zhou Zhang, Haowen Pan, Ran Mao, Zian Zhao, Yier Jin, and Zhenyu Guan. 2023. HE3DB: An Efficient and Elastic Encrypted Database Via Arithmetic-And-Logic Fully Homomorphic Encryption. In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security* (Copenhagen, Denmark) *(CCS '23)*. Association for Computing Machinery, New York, NY, USA, 2930–2944. doi:10.1145/3576915.3616608

[6] Bitcoin Trade History. Accessed 2026. https://www.blockchain.com/charts/trade-volume.

[7] Fabian Boemer, Rosario Cammarota, Daniel Demmler, Thomas Schneider, and Hossein Yalame. 2020. MP2ML: A mixed-protocol machine learning framework for private inference. In *Proceedings of the 15th international conference on availability, reliability and security*. 1–10.

[8] Zvika Brakerski. 2012. Fully homomorphic encryption without modulus switching from classical GapSVP. *Annual cryptology conference* (2012), 868–886.

[9] Zvika Brakerski, Craig Gentry, and Vinod Vaikuntanathan. 2014. (Leveled) fully homomorphic encryption without bootstrapping. *ACM Transactions on Computation Theory (TOCT)* 6, 3 (2014), 1–36.

[10] Jung Hee Cheon, Andrey Kim, Miran Kim, and Yongsoo Song. 2017. Homomorphic Encryption for Arithmetic of Approximate Numbers. *Advances in Cryptology – ASIACRYPT 2017* (2017), 409–437.

[11] Ilaria Chillotti, Nicolas Gama, Mariya Georgieva, and Malika Izabachène. 2018. TFHE: Fast Fully Homomorphic Encryption over the Torus. Cryptology ePrint Archive, Paper 2018/421. https://eprint.iacr.org/2018/421

[12] Covid-19 Data. Accessed 2026. https://covidtracking.com/data/download/national-history.csv.

[13] Saba Eskandarian, Jonathan Cogan, Sawyer Birnbaum, Peh Chang Wei Brandon, Dillon Franke, Forest Fraser, Gaspar Garcia, Eric Gong, Hung T. Nguyen, Taresh K. Sethi, Vishal Subbiah, Michael Backes, Giancarlo Pellegrino, and Dan Boneh. 2019. Fidelius: Protecting User Secrets from Compromised Browsers. In *2019 IEEE Symposium on Security and Privacy (SP)*. 264–280. doi:10.1109/SP.2019.00036

[14] Saba Eskandarian and Matei Zaharia. 2019. ObliDB: oblivious query processing for secure databases. *Proc. VLDB Endow.* 13, 2 (Oct. 2019), 169–183. doi:10.14778/3364324.3364331

[15] Junfeng Fan and Frederik Vercauteren. 2012. Somewhat Practical Fully Homomorphic Encryption. *Cryptology ePrint Archive, Paper 2012/144* (2012). https://eprint.iacr.org/2012/144

[16] Craig Gentry. 2009. Fully homomorphic encryption using ideal lattices. *Proceedings of the forty-first annual ACM symposium on Theory of computing* (2009).

[17] Craig Gentry, Amit Sahai, and Brent Waters. 2013. Homomorphic Encryption from Learning with Errors: Conceptually-Simpler, Asymptotically-Faster, Attribute-Based. In *Advances in Cryptology – CRYPTO 2013*, Ran Canetti and Juan A. Garay (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 75–92.

[18] Antonio Guimarães, Edson Borin, and Diego F. Aranha. 2024. MOSFHET: Optimized Software for FHE over the Torus. *IACR Cryptol. ePrint Arch.* 2022 (2024), 515. https://api.semanticscholar.org/CorpusID:248661190

[19] Hakan Hacigümüş, Bala Iyer, Chen Li, and Sharad Mehrotra. 2002. Executing SQL over Encrypted Data in the Database-Service-Provider Model. In *Proceedings of the 2002 ACM SIGMOD International Conference on Management of Data* (Madison, Wisconsin) *(SIGMOD '02)*. Association for Computing Machinery, New York, NY, USA, 216–227. doi:10.1145/564691.564717

[20] Boyoung Han, Yeonghyeon Kim, Jina Choi, Hojune Shin, and Younho Lee. 2023. Fully Homomorphic Privacy-Preserving Naive Bayes Machine Learning and Classification. In *Proceedings of the 11th Workshop on Encrypted Computing & Applied Homomorphic Cryptography* (Copenhagen, Denmark) *(WAHC '23)*. Association for Computing Machinery, New York, NY, USA, 91–102. doi:10.1145/3605759.3625262

[21] HElib. Accessed 2026. https://github.com/homenc/HElib.

[22] Human Genome Databases. Accessed 2026. http://hgdownload.soe.ucsc.edu/goldenPath/hg38/database/.

[23] Intel Paillier Cryptosystem Library. Accessed 2026. https://github.com/intel/paillier-cryptolib.

[24] Boram Jung, Yuliang Li, and Hung-Wei Tseng. 2026. NSHEDB: Noise-Sensitive Homomorphic Encrypted Database Query Engine. arXiv:2602.24271 [cs.DB] https://arxiv.org/abs/2602.24271

[25] Kate Keahey, Jason Anderson, Zhuo Zhen, Pierre Riteau, Paul Ruth, Dan Stanzione, Mert Cevik, Jacob Colleran, Haryadi S. Gunawi, Cody Hammock, Joe Mambretti, Alexander Barnes, François Halbach, Alex Rocha, and Joe Stubbs. 2020. Lessons Learned from the Chameleon Testbed. In *Proceedings of the 2020 USENIX Annual Technical Conference (USENIX ATC '20)*. USENIX Association.

[26] Lattigo: lattice-based multiparty homomorphic encryption library in Go. Accessed 2026. https://github.com/tuneinsight/lattigo.

[27] Pascal Paillier. 1999. Public-Key Cryptosystems Based on Composite Degree Residuosity Classes. In *Advances in Cryptology — EUROCRYPT '99*, Jacques Stern (Ed.). Springer Berlin Heidelberg, Berlin, Heidelberg, 223–238.

[28] Rishabh Poddar, Tobias Boelter, and Raluca Ada Popa. 2019. Arx: An Encrypted Database using Semantically Secure Encryption. *Proc. VLDB Endow.* 12, 11 (2019), 1664–1678. doi:10.14778/3342263.3342641

[29] Raluca Ada Popa, Catherine Redfield, Nickolai Zeldovich, and Hari Balakrishnan. 2011. CryptDB: protecting confidentiality with encrypted query processing. In *Proceedings of the Twenty-Third ACM Symposium on Operating Systems Principles*. ACM, 85–100.

[30] Savvas Savvides, Darshika Khandelwal, and Patrick Eugster. 2020. Efficient Confidentiality-Preserving Data Analytics over Symmetrically Encrypted Datasets. *Proc. VLDB Endow.* 13, 8 (April 2020), 1290–1303. doi:10.14778/3389133.3389144

[31] SEAL 2021. Microsoft SEAL (release 3.7). https://github.com/Microsoft/SEAL. Microsoft Research, Redmond, WA..

[32] Olamide T. Tawose, Jun Dai, Lei Yang, and Dongfang Zhao. 2023. Toward Efficient Homomorphic Encryption for Outsourced Databases through Parallel Caching. *Proceedings of the ACM on Management of Data (SIGMOD)*, Article 66 (May 2023), 23 pages. doi:10.1145/3588920

[33] Jonathan Takeshita, Dayane Reis, Ting Gong, Michael Niemier, X. Sharon Hu, and Taeho Jung. 2023. Accelerating Finite-Field and Torus FHE via Compute-Enabled (S)RAM. *IEEE Trans. Comput.* (2023), 1–14. doi:10.1109/TC.2023.3301116

[34] Nikolaj Volgushev, Malte Schwarzkopf, Ben Getchell, Mayank Varia, Andrei Lapets, and Azer Bestavros. 2019. Conclave: secure multi-party computation on big data. In *Proceedings of the Fourteenth EuroSys Conference 2019* (Dresden, Germany) *(EuroSys '19)*. Association for Computing Machinery, New York, NY, USA, Article 3, 18 pages. doi:10.1145/3302424.3303982

[35] Yongzhi Wang, Yulong Shen, Cuicui Su, Jiawen Ma, Lingtong Liu, and Xuewen Dong. 2020. CryptSQLite: SQLite With High Data Security. *IEEE Trans. Comput.* 69, 5 (2020), 666–678. doi:10.1109/TC.2019.2963303

[36] Yilei Wang and Ke Yi. 2021. Secure Yannakakis: Join-Aggregate Queries over Private Data. In *Proceedings of the 2021 International Conference on Management of Data* (Virtual Event, China) *(SIGMOD '21)*. Association for Computing Machinery, New York, NY, USA, 1969–1981. doi:10.1145/3448016.3452808

[37] Sai Wu, Feng Li, Sharad Mehrotra, and Beng Chin Ooi. 2011. Query optimization for massively parallel data processing. In *Proceedings of the 2nd ACM Symposium on Cloud Computing* (Cascais, Portugal) *(SOCC '11)*. Association for Computing Machinery, New York, NY, USA, Article 12, 13 pages. doi:10.1145/2038916.2038928

[38] Xun Yi, Md. Golam Kaosar, Russell Paulet, and Elisa Bertino. 2013. Single-Database Private Information Retrieval from Fully Homomorphic Encryption. *IEEE Trans. on Knowl. and Data Eng.* 25, 5 (May 2013), 1125–1134. doi:10.1109/TKDE.2012.90

[39] Shufan Zhang, Xi He, Ashish Kundu, Sharad Mehrotra, and Shantanu Sharma. 2024. Secure Normal Form: Mediation Among Cross Cryptographic Leakages in Encrypted Databases. In *2024 IEEE 40th International Conference on Data Engineering (ICDE)*. 5560–5573. doi:10.1109/ICDE60146.2024.00444

[40] Zhou Zhang, Song Bian, Zian Zhao, Ran Mao, Haoyi Zhou, Jiafeng Hua, Yier Jin, and Zhenyu Guan. 2024. ArcEDB: An Arbitrary-Precision Encrypted Database via (Amortized) Modular Homomorphic Encryption. In *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security* (Salt Lake City, UT, USA) *(CCS '24)*. Association for Computing Machinery, New York, NY, USA, 4613–4627. doi:10.1145/3658644.3670384